

HEALTHCONNECT EXPERIENCE LAB

HC-POD Cross-Track Testing & Validation Evidence

Data Analytics × Data Science | Week 7

Project Information	Details
Analytics author	Nancy Lee YIMBERE ALAPINI — Data Analytics
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Pod	HC-POD 01
Programme	AnalystLab Africa — Premium Data Analytics Internship
Project	HealthConnect Experience Lab
Evidence purpose	Demonstrate meaningful cross-track testing, refinement, retesting and validation — not communication alone.

This document records the Week 7 cross-track validation activity between Data Analytics and Data Science. It demonstrates how validated Analytics findings were converted into explicit modelling questions, tested by Data Science under controlled conditions, and used to make concrete feature-set decisions for Week 8.

CROSS-TRACK CHAIN: DEPENDENCY → TEST → FINDING → ACTION → RETEST → VALIDATED IMPROVEMENT

1. AnalystLab Cross-Track Evidence — 12 Required Elements

#	Required Element	Evidence
1	Track collaborated with	Data Science — HC-POD 01.
2	Dependency	Data Analytics needed Data Science to validate whether Analytics findings influencing model features or modelling decisions added measurable out-of-sample predictive value.
3	Component / output tested	CT01: long_lead_followup. CT02: distance_to_clinic_km.
4	Information received by Data Science	Analytics provided validated evidence: Lead Time 0–30 days NSR 39.13% vs 31–60 days 63.95% (+24.82pp); Previous No-Shows 46.30% vs 57.83%; Distance standalone $p \approx 0.055$; Lead Time × Appointment Type interaction $p \approx 0.326$.
5	Information provided by Data Analytics	Two explicit testing requests: (1) test incremental out-of-sample value of long_lead_followup beyond broader Booking Lead Time; (2) test whether Distance adds incremental multivariable predictive value despite weak standalone Analytics association.
6	Testing activity	Data Science ran controlled with-vs-without feature ablation using the same Logistic Regression pipeline, patient-grouped split and preprocessing. Evaluation included ROC-AUC, grouped 5-fold cross-validation, bootstrap confidence intervals and Top-20%-risk business lift where relevant.
7	Issue / finding	CT01: no statistically meaningful incremental value for long_lead_followup. CT02: Distance showed a small but consistent multivariable contribution despite weak standalone association.
8	Refinement / action	Remove long_lead_followup from the Week 8 candidate feature set. Retain distance_to_clinic_km in the predictive model while keeping its Analytics classification Secondary / Contextual.
9	Retest result	CT01 performance remained statistically indistinguishable without the engineered feature. CT02 grouped CV favoured the model with Distance in 5/5 folds; business lift also weakened when Distance was removed.
10	What changed	The Week 8 candidate feature set became simpler and more defensible. Distance retention became evidence-based rather than assumed. Analytics and Data Science interpretations were explicitly separated.
11	Evidence	Data Science cross-track validation notebook containing ablation code, bootstrap results, grouped 5-fold CV, business-lift comparison and fold-level outputs; Week 7 cross-track evidence document; Analytics T10/T11 validation results.
12	How this improved HealthConnect	The collaboration removed an engineered feature without measurable loss, validated a modest multivariable contributor, reduced overinterpretation risk and strengthened the evidence chain from Analytics to predictive modelling and Week 8 integration.

2. Analytics Evidence Shared with Data Science

The cross-track request was initiated only after the Data Analytics workstream completed its own Week 7 validation. The following results were therefore treated as the validated analytical reference:

Analytics Result	Validated Evidence	Interpretation
Booking Lead Time	0–30 days: 39.13% NSR; 31–60 days: 63.95%; gap +24.82pp	PRIMARY standalone analytical differentiation
Recorded Previous No-Shows	0: 46.30% NSR; ≥1: 57.83%; gap +11.53pp	SUPPORTING / COMPLEMENTARY
Distance to Clinic	Spearman $p \approx 0.055$	Very weak standalone association; SECONDARY / CONTEXTUAL
Lead Time × Appointment Type	Follow-up descriptive gap +28.73pp; interaction $p \approx 0.326$	No supported distinct interaction; no Follow-up-specific rule

3. CT01 — Incremental Value of long_lead_followup

3.1 Testing question

Does the engineered long_lead_followup feature provide measurable incremental out-of-sample predictive value beyond the broader Booking Lead Time signal?

3.2 Why this test mattered

Week 6 model outputs had assigned importance to the engineered feature. However, Week 7 Analytics robustness testing showed that the Lead-Time pattern persisted across appointment types while the formal Lead Time × Appointment Type interaction was not statistically supported. Feature importance alone was therefore insufficient to justify retaining the engineered feature.

3.3 Data Science test

- With-vs-without feature ablation under the same Logistic Regression pipeline.
- Same patient-grouped split and preprocessing.
- Single-split ROC-AUC difference.
- Bootstrap 95% confidence interval for the difference.
- Grouped 5-fold cross-validation.

3.4 Result

Validation view	Result
Single-split AUC difference	+0.0017
Bootstrap 95% CI	[-0.0008, 0.0041] — includes zero
Grouped 5-fold CV mean difference	+0.0009
Overall interpretation	No statistically meaningful incremental predictive value demonstrated

3.5 Action and retest

The feature was removed from the Week 8 candidate feature set. The model without long_lead_followup remained statistically indistinguishable in predictive performance and showed no evidence of material degradation.

CT01 — PASS | TESTED → REFINED → RETESTED → VALIDATED | FEATURE REMOVED

4. CT02 — Incremental Value of distance_to_clinic_km

4.1 Testing question

Can Distance to Clinic provide incremental multivariable predictive value even though its standalone Analytics association with No-Show behaviour is very weak?

4.2 Why this test mattered

Analytics had correctly classified Distance as Secondary / Contextual because its standalone association was weak. Data Science, however, evaluates the joint predictive contribution of features. The test therefore examined whether weak standalone differentiation necessarily implied no value inside a multivariable model.

4.3 Result

Validation view	Result
Analytics standalone association	Spearman $\rho \approx 0.055$ — very weak
Single-split bootstrap CI	[-0.0089, 0.0096] — inconclusive
Grouped 5-fold CV	Model with Distance outperformed model without Distance in 5/5 folds
Mean grouped-CV difference	+0.0043
Top-20%-risk business lift	1.44× with Distance vs 1.40× after removal

4.4 Action and retest

Distance was retained in the predictive model as a modest but consistent multivariable contributor. This does not change its Data Analytics classification and does not justify standalone operational targeting.

CT02 — PASS | TESTED → RETESTED → RETAINED

5. Cross-Track Interpretation

The two tests answer different questions from standalone descriptive Analytics. CT01 showed that a model feature can appear important in one model output without adding unique predictive value once controlled through ablation. CT02 showed that a feature with weak standalone association can still contribute modestly within a multivariable model.

Standalone analytical differentiation ≠ Multivariable predictive contribution ≠ Causality

6. Before / After Cross-Track Improvement

Component	Before Cross-Track Test	After Testing / Refinement	Week 8 Impact
long_lead_followup	Candidate engineered feature; model importance did not establish unique value	No meaningful incremental OOS value; feature removed	Simpler candidate feature set without measurable loss
distance_to_clinic_km	Weak standalone Analytics association; predictive contribution uncertain	Small but consistent multivariable contribution; retained	Retention supported by ablation/CV evidence
Analytics ↔ DS interpretation	Potential risk of treating model importance as business importance	Evidence types explicitly separated	Safer integrated storytelling and decision support

7. Validation Outcome & Contribution to HealthConnect

- The cross-track activity tested real modelling questions derived from validated Analytics findings.
- The collaboration produced concrete feature-set changes rather than communication-only evidence.
- The final model candidate set is simpler because long_lead_followup was removed without demonstrated loss.
- Distance retention is now supported by multivariable validation rather than assumption.
- The Analytics evidence hierarchy remains unchanged: Booking Lead Time PRIMARY; Recorded Previous No-Shows SUPPORTING / COMPLEMENTARY; Reminder Status and Distance SECONDARY / CONTEXTUAL.
- No new causal claim or standalone Distance targeting rule was introduced.
- No unresolved Week 7 Data Analytics → Data Science testing dependency remains.

8. Evidence Traceability

Evidence Item	Purpose / Traceability
Week 7 Analytics Testing & Validation Record	Documents T01–T14, including T10 Distance and T11 Appointment Type × Lead Time.
HC-POD Cross-Track Testing & Validation Evidence	Documents dependency, test design, findings, actions, retests and validated changes.
Data Science validation notebook	Contains controlled ablation code, bootstrap results, grouped CV and business-lift outputs.
Week 7 Main Analytics Report	Integrates the validated cross-track interpretation into the final Analytics narrative.
Week 7 Project Summary	Records the collaborative test, what changed and Week 8 readiness.

9. Week 8 Handoff

The cross-track testing dependency is closed for Week 7. Week 8 integration should carry forward the refined candidate feature set and preserve the distinction between Analytics evidence and predictive model contribution. Any attendance-support intervention must remain under a Test & Measure framework because neither the Analytics findings nor predictive model contributions establish causal intervention effectiveness.

FINAL STATUS: MEANINGFUL CROSS-TRACK TESTING COMPLETED | DEPENDENCIES RESOLVED | READY FOR WEEK 8 INTEGRATION